LINA - Lab04 – Functions (trig+inverse)

Work individually and make sure that throughout the course of this lab, you re-save sufficiently your solutions file on your own machine, finally delivered .**PDF**

LINA_2627_IGP99E_Van_Der_Zever_Johnny_Lab**04**.**PDF** (< 5 MB),

Reduce via https://www.ilovepdf.com/compress_pdf or Adobe Acrobat (Online)

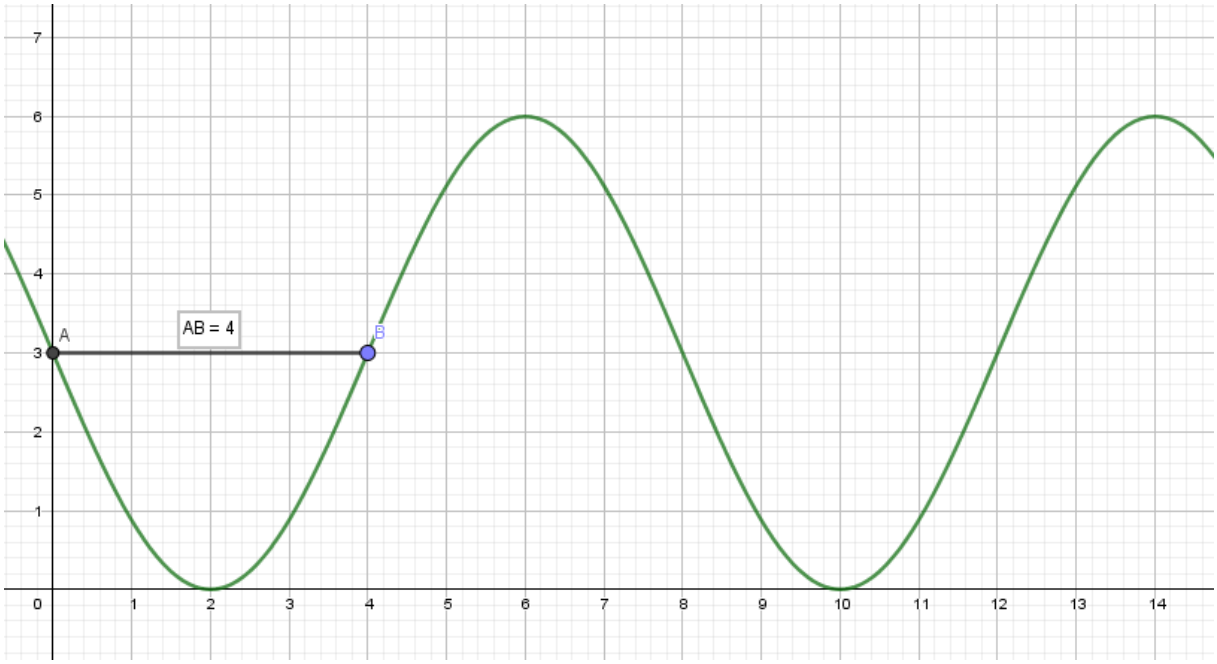
➔ which is to submit via week-folder Leho/ LINA / Week**04**/Lab**04** Practice

Also, make sure to repeat Class group + Full Name +Lab**04** number inside it.

1. Exercises

1.1. Basic exercise

1.1.1. Retrieve the sine function



Retrieve all coefficients by the graph of this pictured general sine function. Provide for each found coefficient a short but concise reasoning

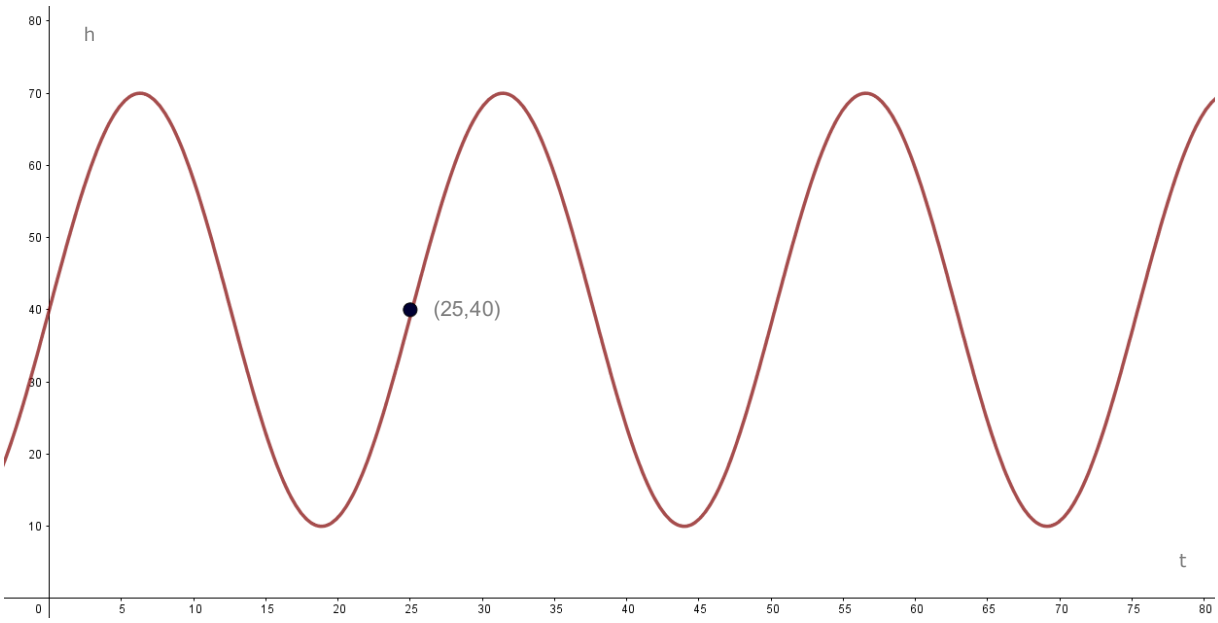
$$f(x) = r\sin(\omega x + \theta_0) + c$$

- $r =$
- $\omega =$
- $\theta_0 =$
- $c =$

1.2. Bridging exercise

1.2.1. Hovering Motion

This sine-graph describes the hovering motion of a hovercraft when in idle state. The **height h** is plotted against the **time t**. Retrieve all four coefficients.



1. Pulsation =
2. Amplitude =
3. Intercept =
4. Phase =

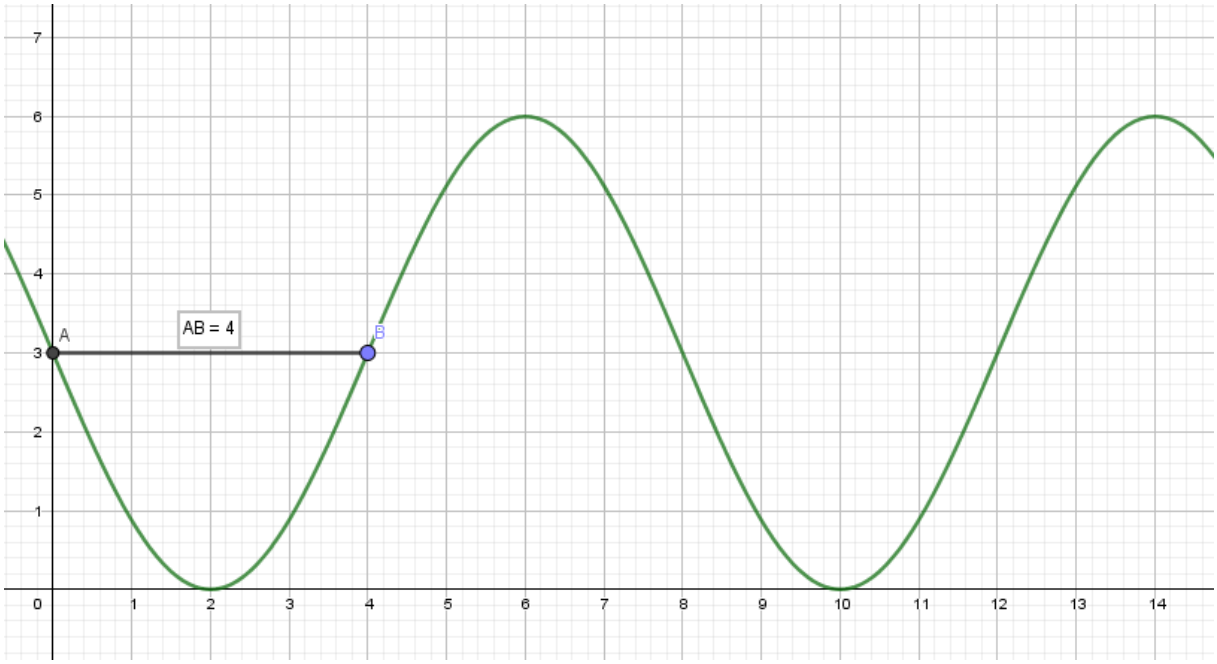
- Write down the recipe for $h(t)$ which is represented by the graph above.

- Estimate at what time t the hovercraft is closest to the ground for the first time.

- Give a mathematical expression for the time t the hovercraft is closest to the ground for the first time.

1.3. Basic exercise (revisited)

1.3.1. Reproduce the sine function's graph

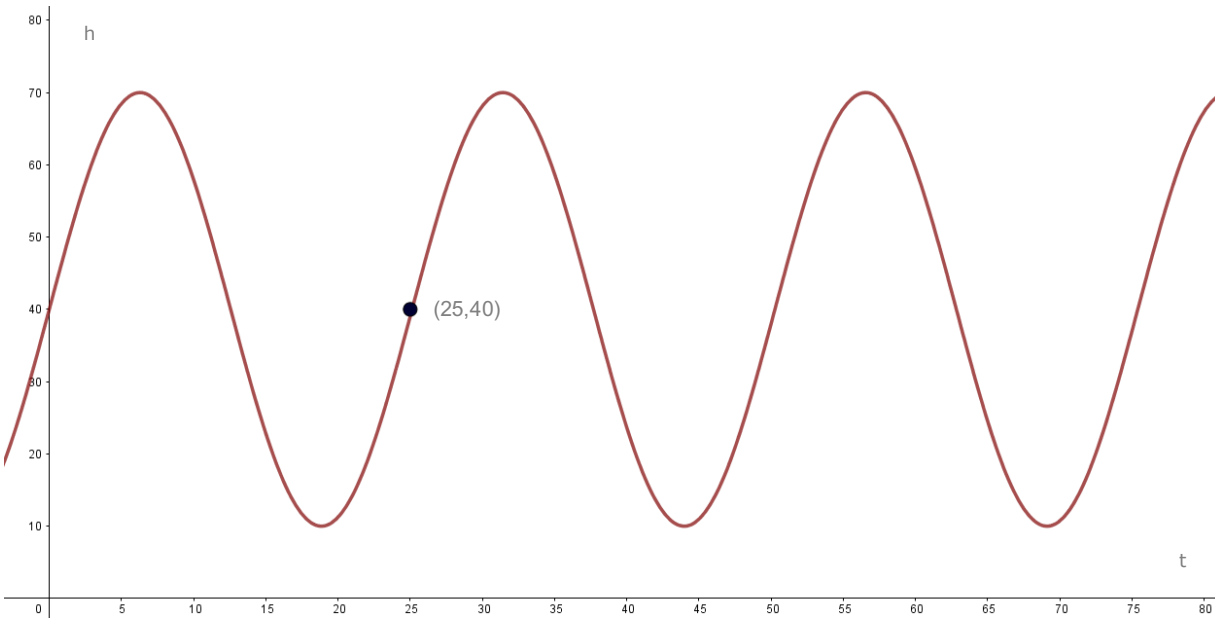


Based on your previously retrieved general sine $f(x) = r\sin(\omega x + \theta_0) + c$

Make this function in GeoGebra and reproduce its graph, to verify your finding.

1.4. Bridging exercise (revisited)

1.4.1. Reproduce the Hovering Motion's graph



Based on your previously retrieved height function $h(t) = r\sin(\omega t + \theta_0) + c$

Make this function in GeoGebra and reproduce its graph, to verify your finding.

We continue with two extra challenges on the hover craft's height function.

- Make a point (slider) in GeoGebra that describes the *motion* of the hovercraft. The height is plotted on the y axis and the horizontal position on the x axis stays at 0. This should result in a vertical up and down *motion* accordingly.

- If your observation starts at t=0, at what time is the hovercraft closest to the ground for the first time? Estimate (by sliding) and then calculate the exact corresponding expression.

1.5. Contextual practice

1.5.1. Moving Platforms

Find the sine waves and points representing the following platforms. Assume that the motion is anchored in the top left corner of the platform. All platforms are 1 unit high and 3 units wide. Make a representation of all platforms in GeoGebra.

- 1st platform moves between (2,5) and (12,5) with a pulsation of 2 Pi /10 Start with a pen and paper sketch to prepare your work in GeoGebra.

- 2nd platform moves between (3,17) and (3,6) with a pulsation of 2 Pi

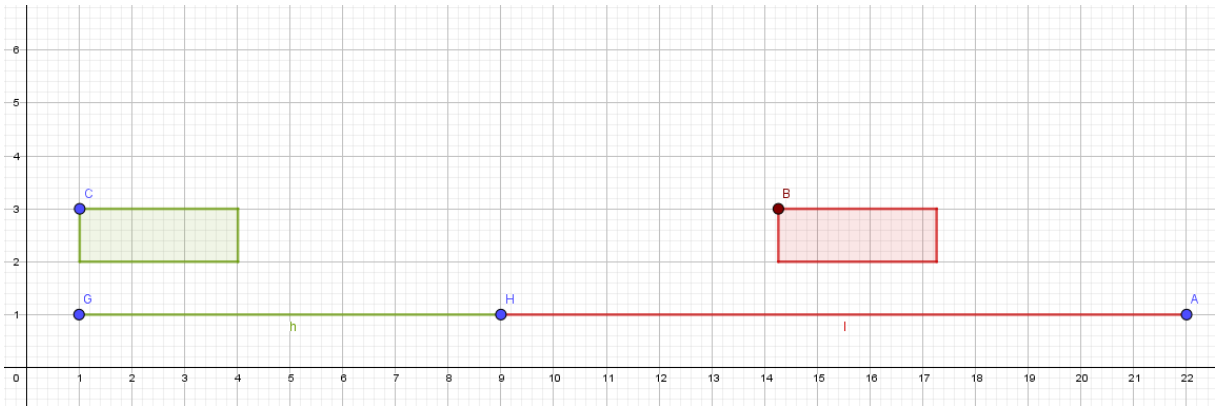
- 3rd platform moves diagonally from (2, 4) to (10, 12). It takes the platform 15 seconds to go from the first point to the second and back.

  Be aware: two individual motions run simultaneously to create the described motion in the diagonal direction.

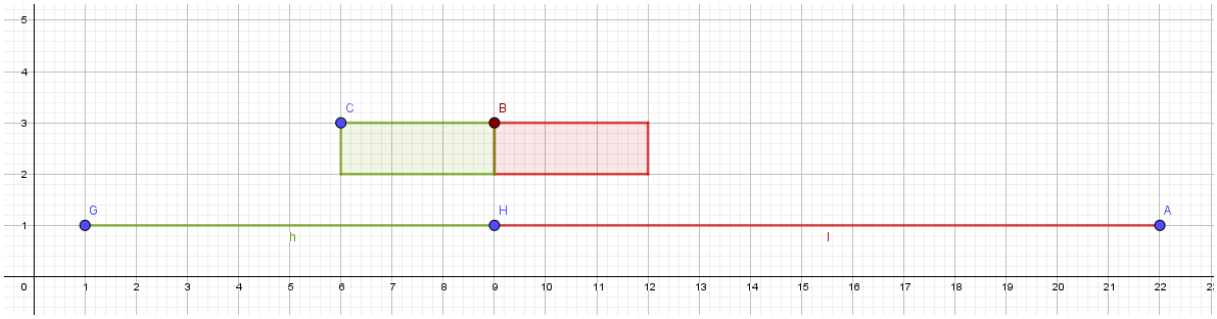
- Configure the slider t from 0 to the period T (set Repeat to: 'Increasing') to obtain a smooth endless loop.

1.5.2. Advanced Platformer Puzzle

For a Mario like platformer you want to create the following platform movement:



The distances under the platforms show the range that the corresponding platforms move. (Including the width of the platform: the point C on the green platform never moves further than 6)



We request a motion by slider, according to these configurations:

- The initial situation is at time t=0 as shown.
- Every 2nd move of the left platform, the platforms touch each other always at the same spot above H.
- It takes the green platform 1.5 seconds to go from left to right. While the red platform takes twice as long.
- The platforms have no vertical movement: the top points of both platforms stay at height 3.
- Both platforms are again 3 units wide and 1 unit high.

Advised way of working:

a. First make your own pen and paper sketch to capture the problem. Also implement all of the specifications into the sketch.

b. Tune both general sine functions to achieve the described animation.

c. Now simulate this entire animated scene in GeoGebra.

1.5.3. Inverse trigonometric functions

a. Inverse Sine function

- Plot in GeoGebra the $arcsin(x)$ function
- Determine its (restricted) domain and according range. List its root(s).

b. Inverse Cosine function

- Plot in GeoGebra the $arccos(x)$ function
- Determine its (restricted) domain and according range. List its root(s).

c. Inverse Tangent function

- Plot in GeoGebra the (restricted) mathematical $arctan(x)$ function
- Determine its (restricted) domain and according range. List its root(s).

d. The programmer's atan2(y,x) **function in GeoGebra**

In GeoGebra its domain takes all (y,x) and its range is $]-\pi, \pi]$ all angles:

https://www.geogebra.org/m/wtkTa3Gw

- Compute in GeoGebra the $atan2d(1,\ 1)$ return
- Compute in GeoGebra the $atan2d(1, -1)$ return
- Compute in GeoGebra the $atan2d(-1,\ 1)$ return
- Compute in GeoGebra the $atan2d(-1, -1)$ return
- Compute in GeoGebra the $atan2d(\ 0, -1)$ return
- Draw in GeoGebra's 3DGraphics the function $z = atan2d(x, y)$
  Explain in your owns words why the vertical stretch occurs in its graph.